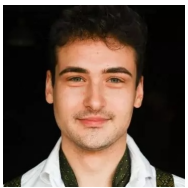
Minimax Optimal Early-Stopped Gradient Descent for Gaussian Mixture Classification

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Joint work with:



Alex Buna



Patrick Rebeschini

Statistics Seminar
School of Mathematics, University of Bristol

30 Oct 2026

Training samples: $(\boldsymbol{X}_i, Y_i)_{i=1}^n$ are i.i.d. copies of $(\boldsymbol{X}, Y) \in \mathbb{R}^d \times \mathbb{R}$
Let $\mathbf{X} := [\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n]^\top \in \mathbb{R}^{n \times d}$, $\boldsymbol{Y} := [Y_1, \ldots, Y_n]^\top \in \mathbb{R}^n$

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GD on empirical risk: $\boldsymbol{w}_0 = \mathbf{0}$, $\boldsymbol{w}_{t+1} = \boldsymbol{w}_t - \gamma \nabla \widehat{\mathcal{L}}(\boldsymbol{w}_t)$ for $t \geq 0$ with step size γ

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Least-squares regression **Linear classification**

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Observation	$Y \in \mathbb{R}$	$Y \in \{\pm 1\}$

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Objective	$\widehat{\mathcal{L}}(\boldsymbol{w}) = \frac{1}{2n} \|\boldsymbol{Y} - \mathbf{X}\boldsymbol{w}\|^2$	$\widehat{\mathcal{L}}(\boldsymbol{w}) = -\frac{1}{n} \sum_{i=1}^n \log \sigma\big(Y_i \boldsymbol{X}_i^\top \boldsymbol{w}\big)$
		$\sigma(z) = 1/(1+e^{-z})$

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GD iterate	$\boldsymbol{w}_t = \big(\boldsymbol{I} - (\boldsymbol{I} - \frac{\gamma}{n}\mathbf{X}^\top\mathbf{X})^t\big)(\mathbf{X}^\top\mathbf{X})^\dagger \mathbf{X}^\top \boldsymbol{Y}$	$\boldsymbol{w}_{t+1} = \boldsymbol{w}_t + \frac{\gamma}{n} \sum_{i=1}^n \sigma(-Y_i \boldsymbol{X}_i^\top \boldsymbol{w}_t) Y_i \boldsymbol{X}_i$

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As $t \to \infty$	$\boldsymbol{w}_t \to (\mathbf{X}^\top\mathbf{X})^\dagger \mathbf{X}^\top \boldsymbol{Y}$, min-$\ell_2$-norm interpolator	$\|\boldsymbol{w}_t\| \to \infty, \boldsymbol{w}_t/\|\boldsymbol{w}_t\| \to \boldsymbol{w}_{\mathsf{MM}}/\|\boldsymbol{w}_{\mathsf{MM}}\|$ $\boldsymbol{w}_{\mathsf{MM}}$ is max-margin classifier

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Target risk	$\mathbb{E}\big[(Y - \boldsymbol{X}^\top \boldsymbol{w})^2\big]$	$\mathbb{P}\big(Y \neq \operatorname{sign}(\boldsymbol{X}^\top \boldsymbol{w})\big)$

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Target risk	$\mathbb{E}\big[(Y - \boldsymbol{X}^\top \boldsymbol{w})^2\big]$	$\mathbb{P}\big(Y \neq \operatorname{sign}(\boldsymbol{X}^\top \boldsymbol{w})\big)$
Literature	Bartlett et al. [2020], Belkin et al. [2020], Hastie et al. [2022], Tsigler and Bartlett [2023], Muthukumar et al. [2020], Bühlmann and Yu [2003], Raskutti et al. [2014],...	Soudry et al. [2018], Muthukumar et al. [2021], Hsu et al. [2021], Chatterji and Long [2021], Wu et al. [2025b],...

A canonical classification task

A canonical classification task

Figure: Gaussian mixture with $d = 40$, $n = 22$.

A canonical classification task

- Latent label $\widetilde{Y} \sim$ uniform$\{\pm 1\}$

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• $\widetilde{Y} = -1$ • $\widetilde{Y} = 1$

A canonical classification task

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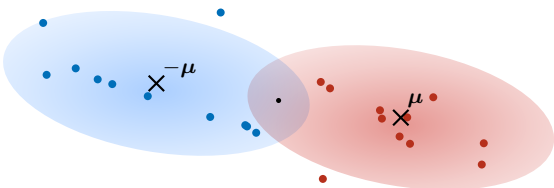


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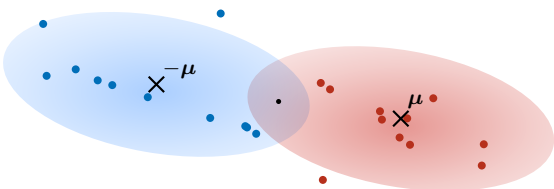


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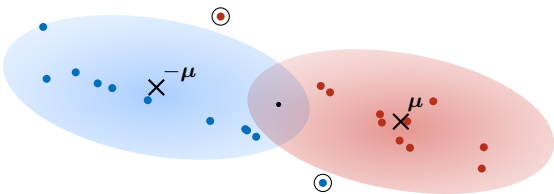
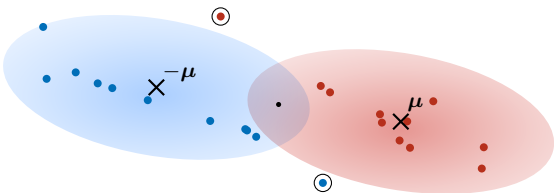


Figure: Gaussian mixture with $d=40$, $n=22$, $p=0.1$.

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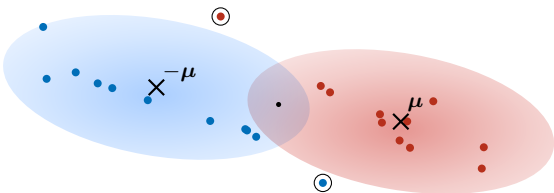
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Regime: $d \geq n$

Figure: Gaussian mixture with $d = 40$, $n = 22$, $p = 0.1$.

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Regime: $d \geq n \Rightarrow$ samples are linearly separable

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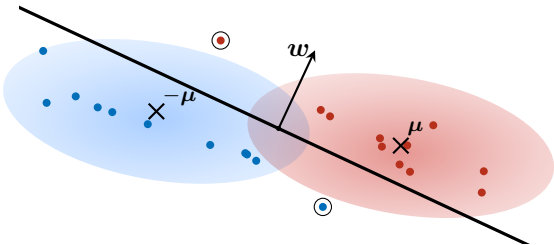


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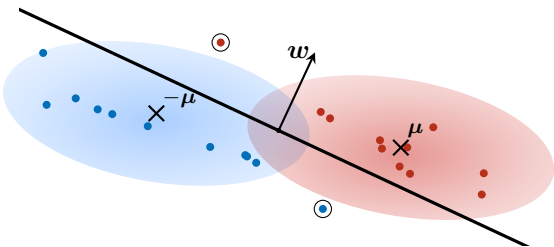


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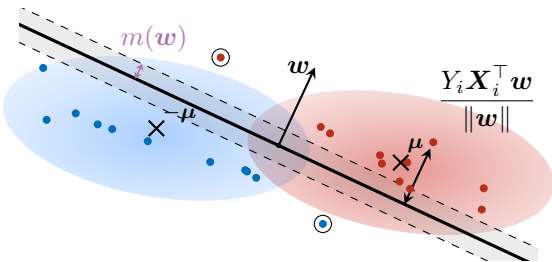


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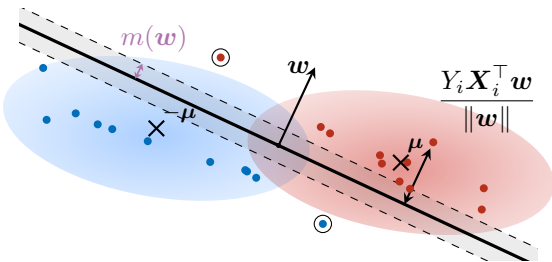


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$\boldsymbol{w}_{\mathsf{MM}} \in \arg\max_{\boldsymbol{w} \in \mathbb{R}^d} \; m(\boldsymbol{w})$

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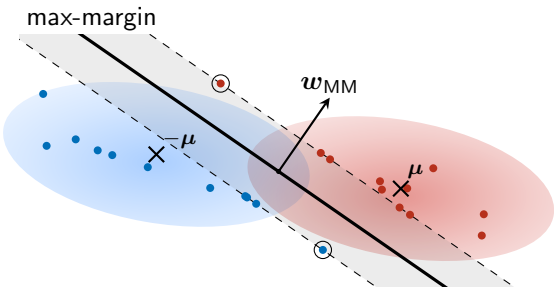


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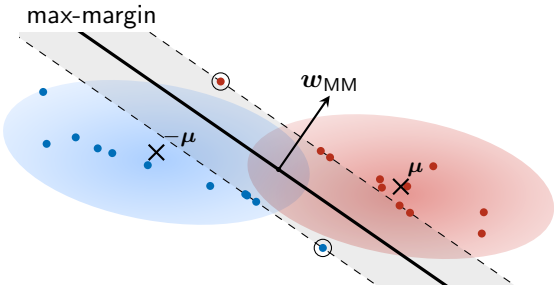


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Regime: $d \geq n \Rightarrow$ samples are linearly separable

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Max-margin classifier:
$\boldsymbol{w}_{\mathsf{MM}} \in \arg\max_{\boldsymbol{w}\in\mathbb{R}^d} \; m(\boldsymbol{w})$

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A canonical classification task

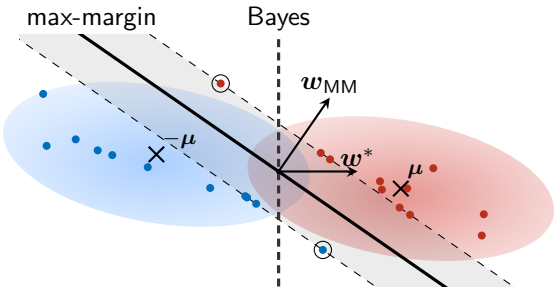


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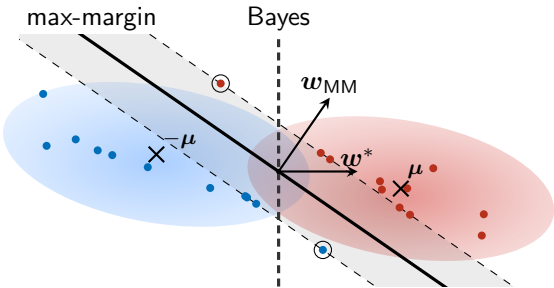


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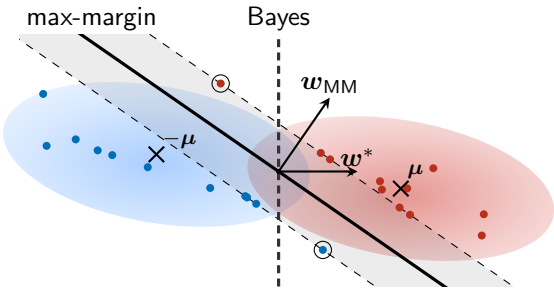


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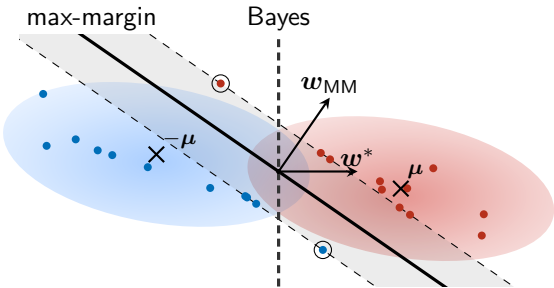


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Soudry et al. [2018], Shamir [2021]

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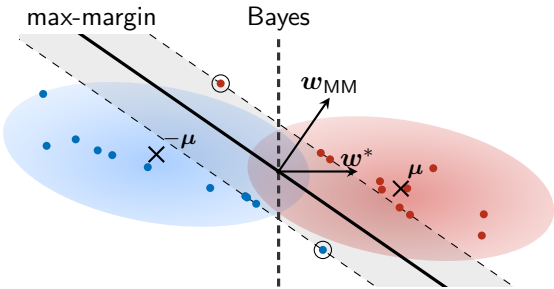


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Goal. Study how the excess zero-one risk

$$\mathcal{E}(\boldsymbol{w}_t) - \mathcal{E}(\boldsymbol{w}^*)$$

evolves along the GD trajectory, where

$$\mathcal{E}(\boldsymbol{w}) := \mathbb{P}\big(Y \neq \mathrm{sign}(\boldsymbol{X}^\top \boldsymbol{w})\big).$$

GD trajectory in practice

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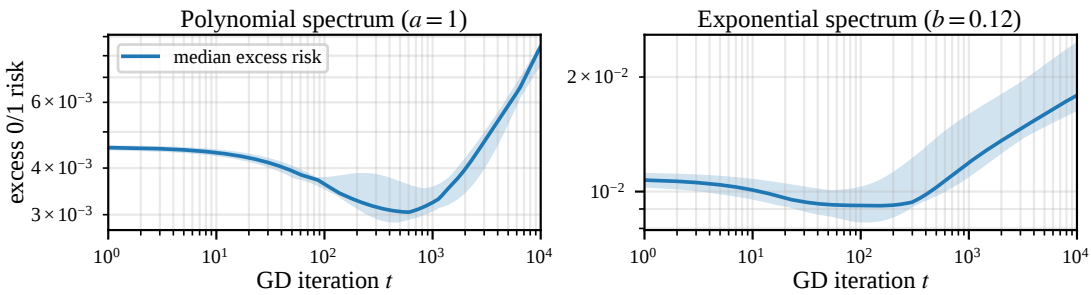
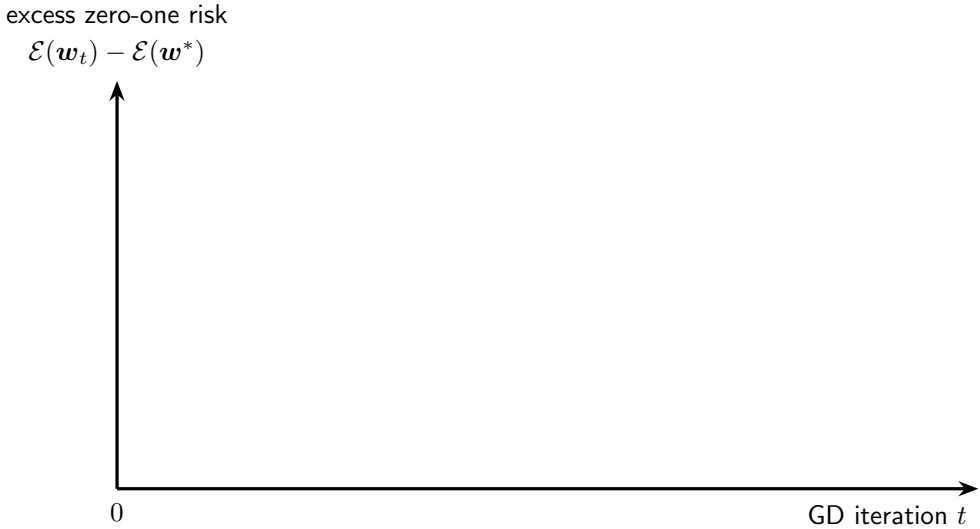


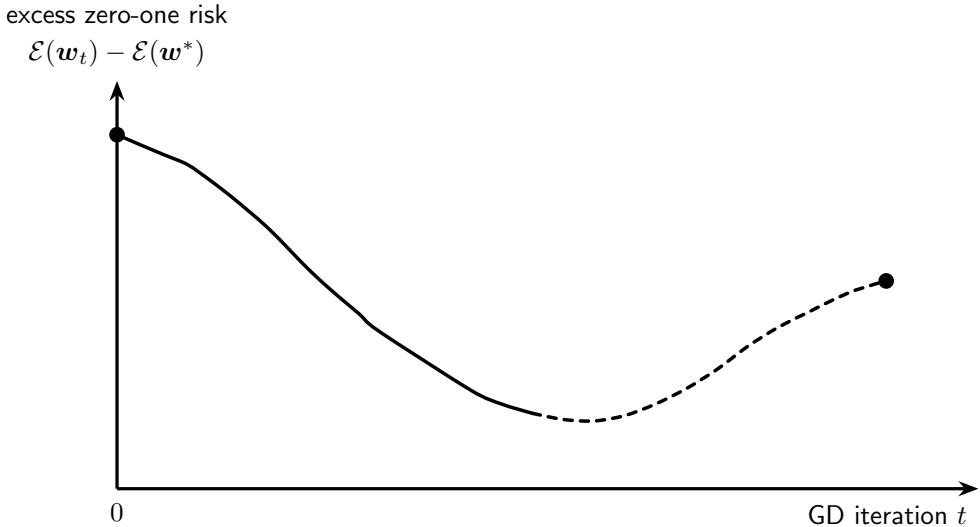
Figure: Excess zero-one risk along GD trajectory. Gaussian mixture with $d = 3888$, $n = 972$. Left: polynomial spectrum $\lambda_i(\mathbf{\Sigma}) \asymp i^{-2a}$. Right: exponential spectrum $\lambda_i(\mathbf{\Sigma}) \asymp e^{-bi}$.

The schematic U curve

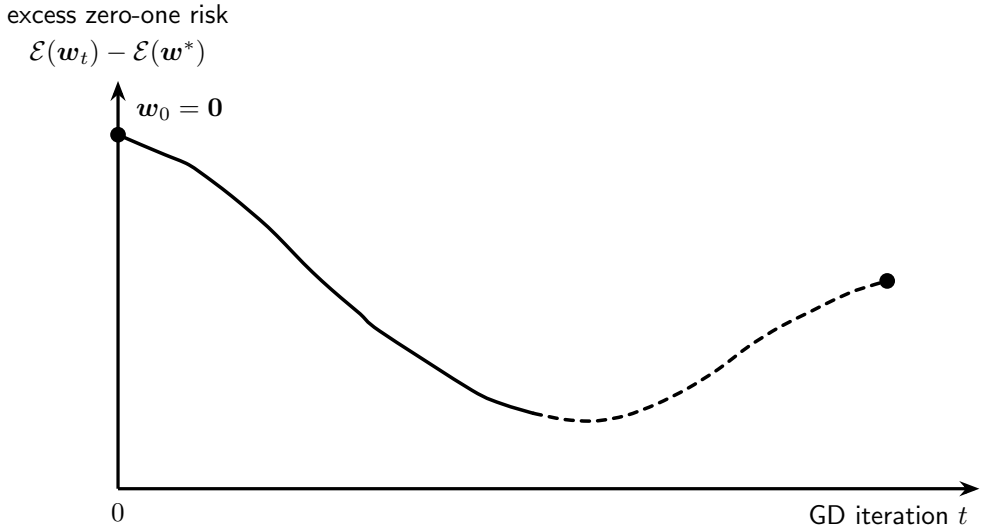
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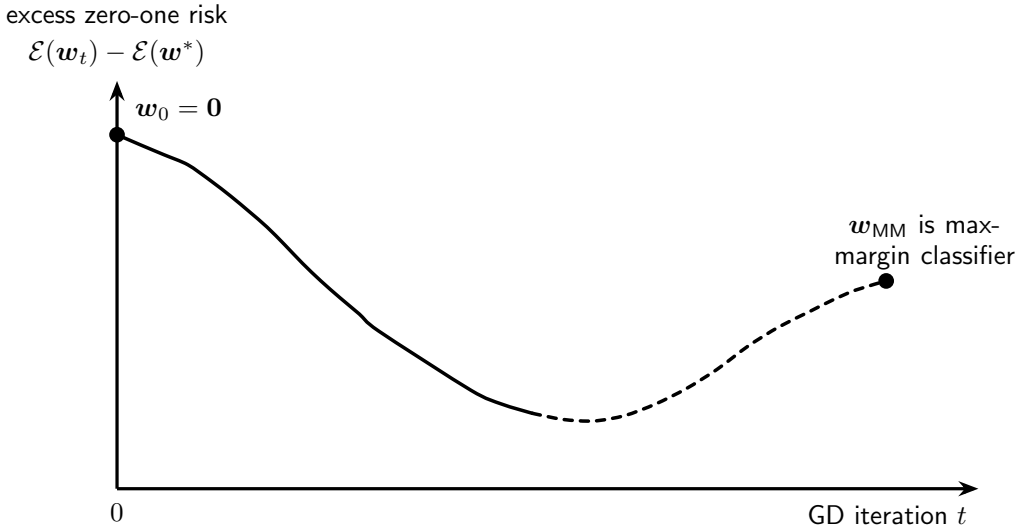
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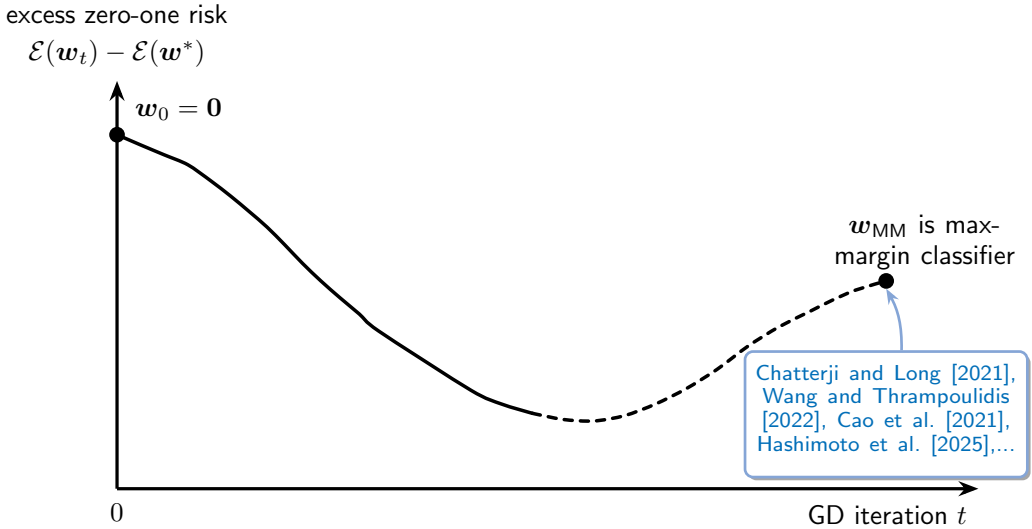
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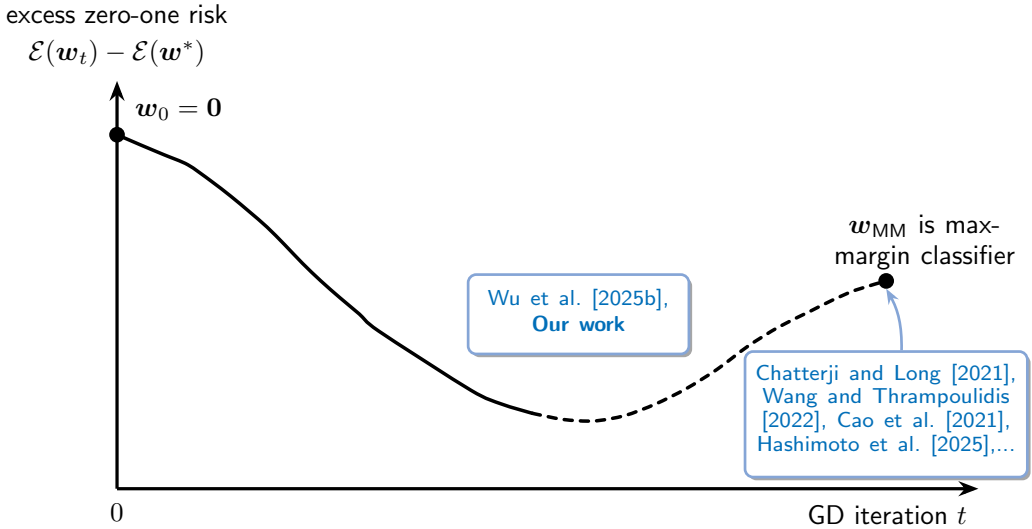
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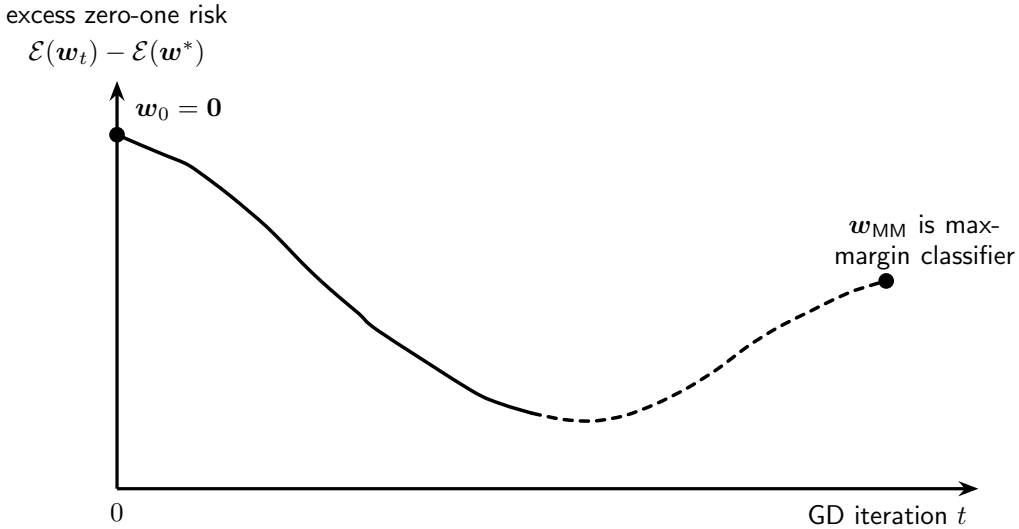
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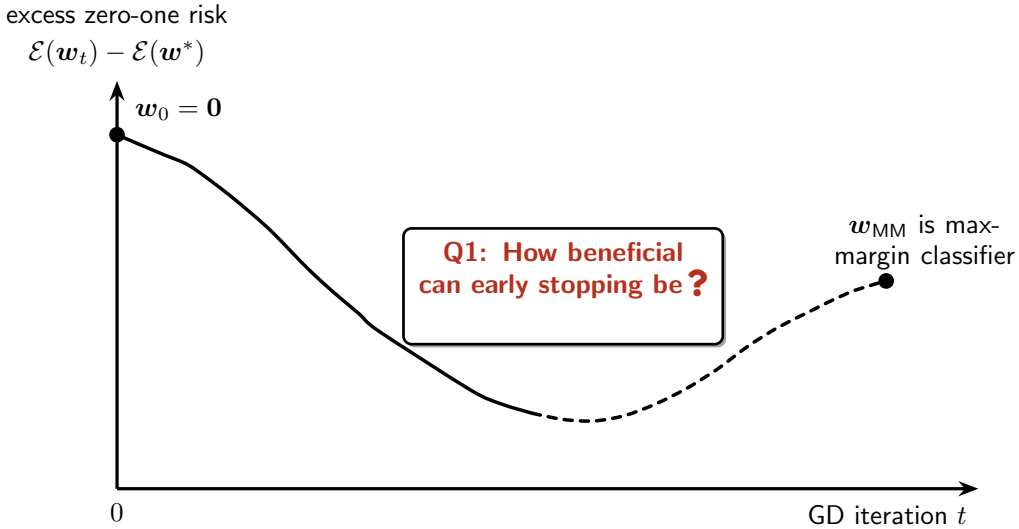
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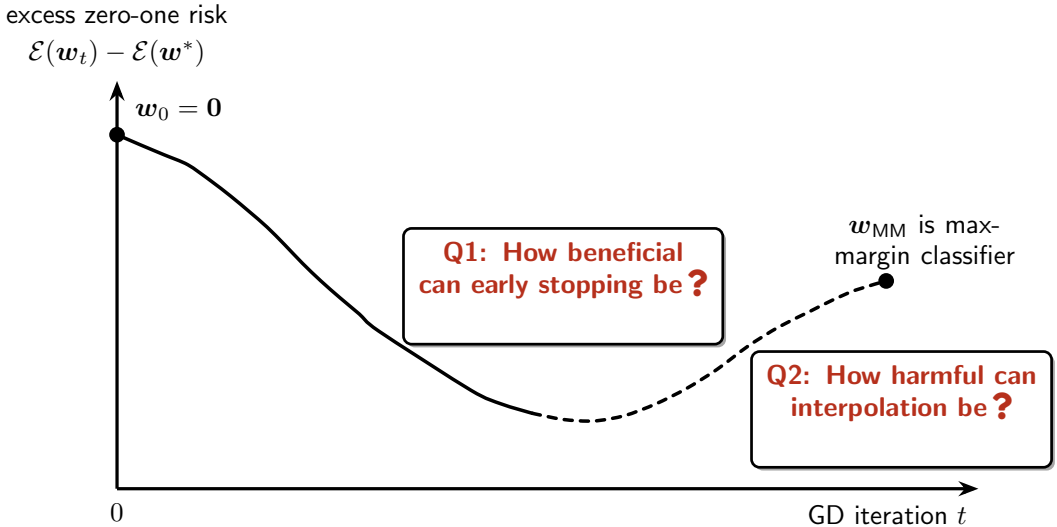
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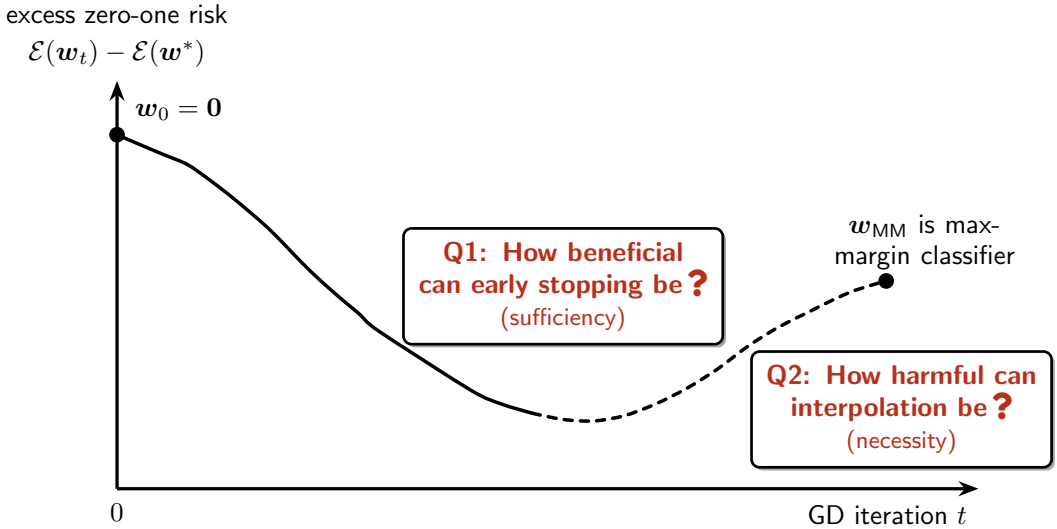
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Part 1: Statistical optimality of early stopping

Where to stop?

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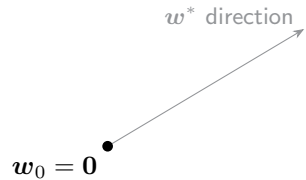


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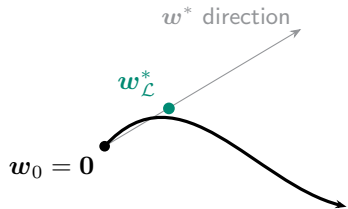


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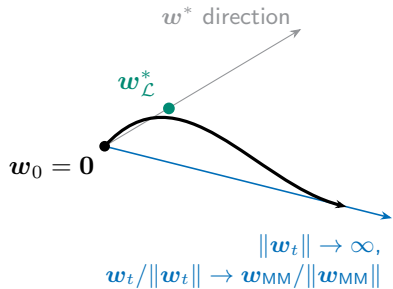


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Target of early stopping: come close to $\boldsymbol{w}^*_{\mathcal{L}}$, then stop before the norm grows.

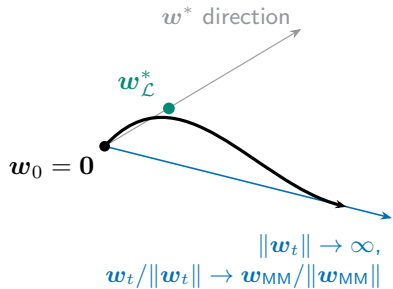


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Proposed stopping rule

[A. Buna, S. X. Liu, P. Rebeschini 2026]

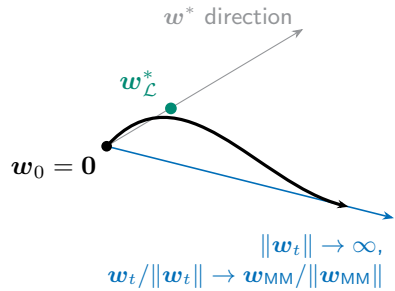


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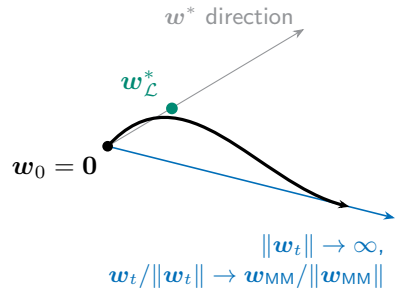


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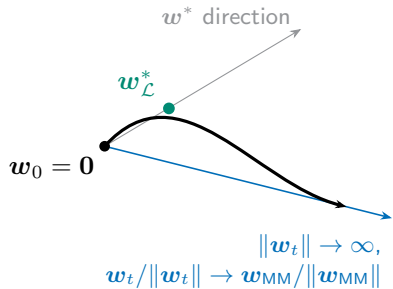


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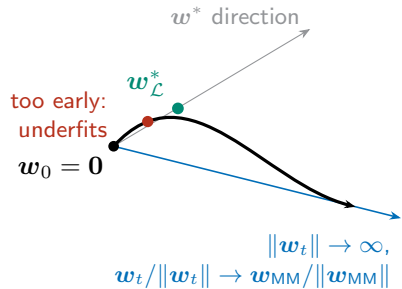


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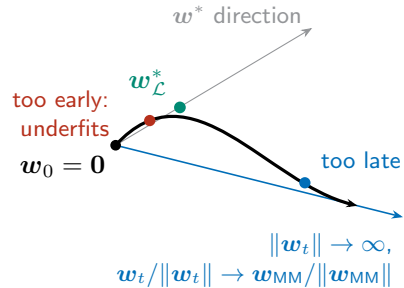


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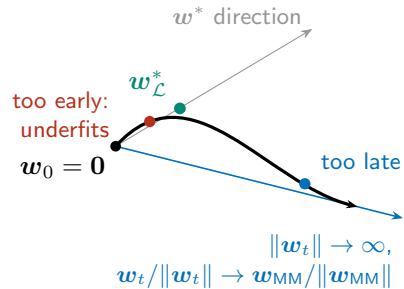


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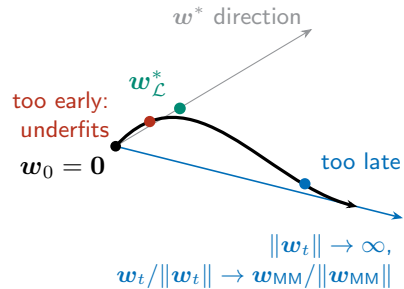


Figure: Schematic GD path in the parameter $\boldsymbol{w}$ space.

Proposed stopping rule

[A. Buna, S. X. Liu, P. Rebeschini 2026]

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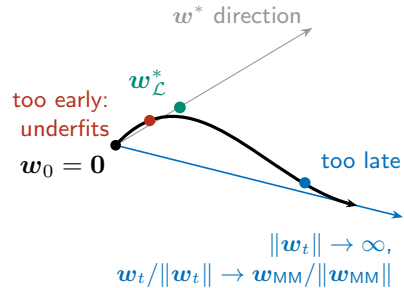


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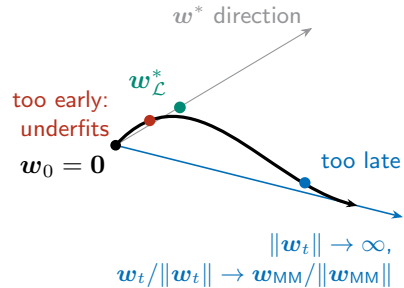


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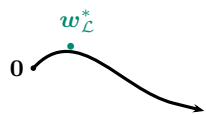
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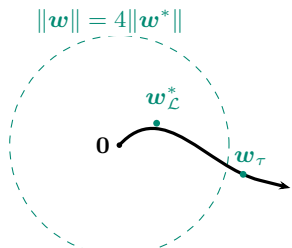


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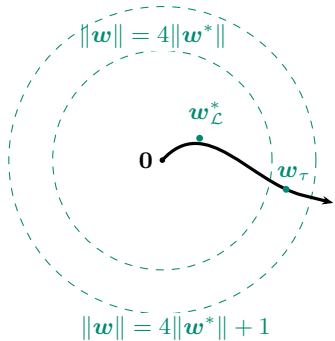


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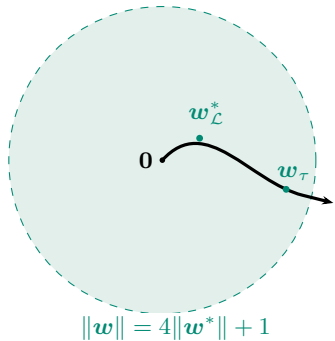
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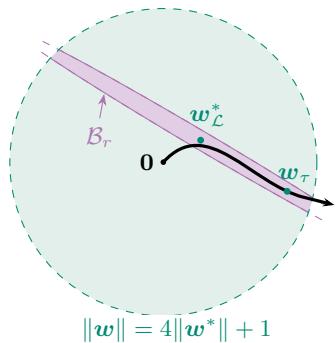
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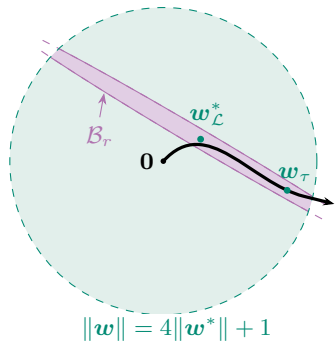
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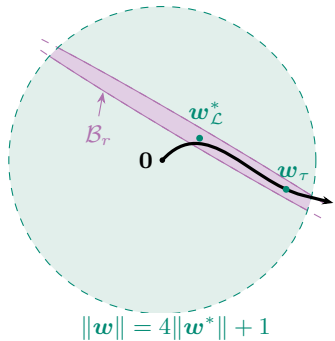
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- The fluctuation over $\mathcal{B}_r$ is controlled by how well $\boldsymbol{h}$ can correlate with noise (Rademacher complexity).



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Offset. For every $\boldsymbol{h}$, subtract and add a quadratic, then take suprema:

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Since the noise has covariance $\boldsymbol{\Gamma}/n$,

$$\mathbb{E}\big[\text{noise term}\big] \asymp \frac{\operatorname{trace}\big(\boldsymbol{\Gamma}(\boldsymbol{\Gamma}+\eta\boldsymbol{I})^{-1}\big)}{n} \lesssim \frac{r_{\text{eff}}(\eta)}{n}, \qquad r_{\text{eff}}(\eta) := \sum_{i=1}^d \frac{\lambda_i}{\lambda_i+\eta}.$$

Combining all components

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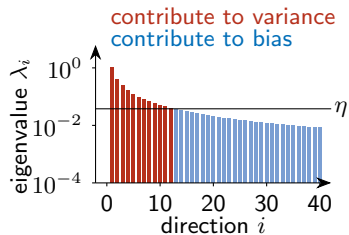
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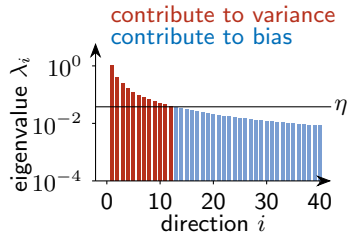
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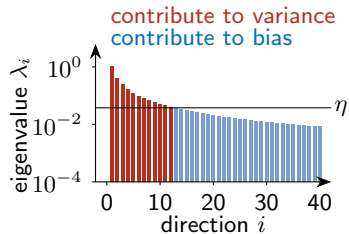
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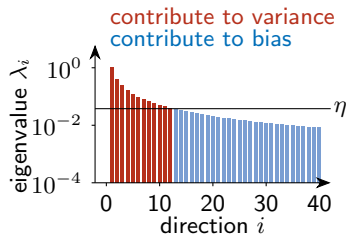
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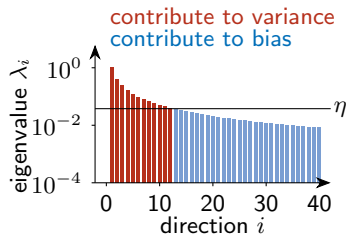


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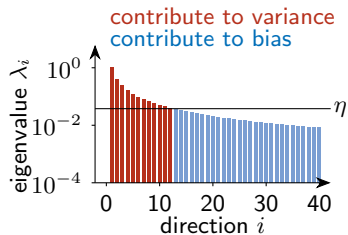
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Fast decaying spectrum	$\mathcal{E}(\boldsymbol{w}) - \mathcal{E}(\boldsymbol{w}^*)$
$\lambda_i \asymp i^{-2a}$, $a > \frac{1}{2}$	$\asymp n^{-\frac{2a}{2a+1}}$
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Table: Both rates match the minimax lower bound.



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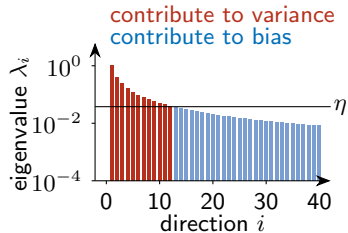
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For flat spectrum, e.g., $\boldsymbol{\Sigma} = \boldsymbol{I}_d$, RHS $= \frac{1}{n} \cdot \frac{d}{1+\eta} + \eta$, recovering classical $\frac{d}{n}$ rate as $\eta \to 0$.

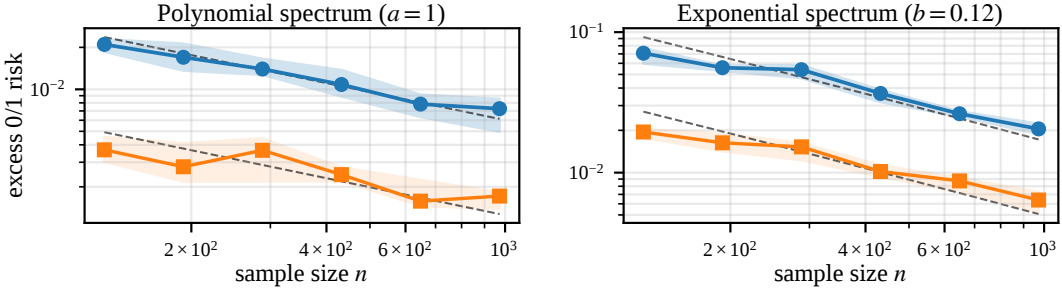


Theory is validated by experiments

[A. Buna, S. X. Liu, P. Rebeschini 2026]

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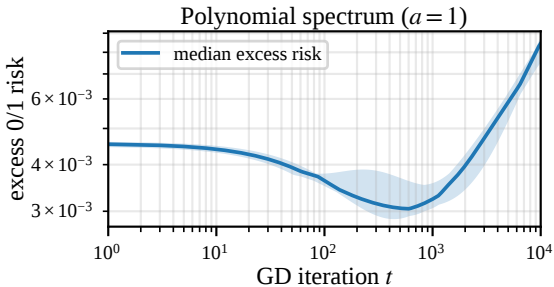
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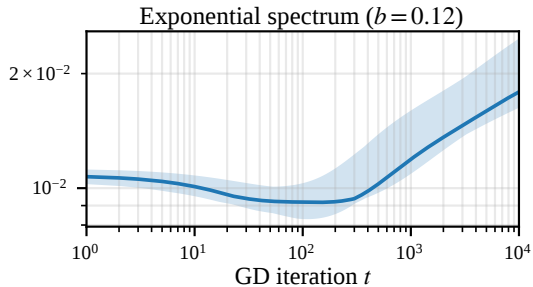


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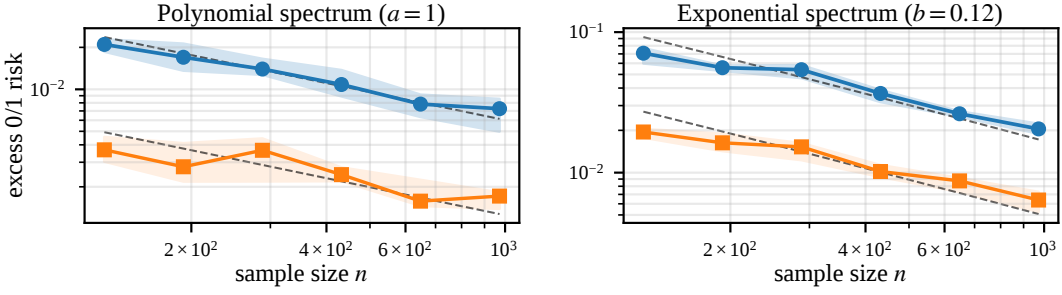
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Part 2: Statistical suboptimality of interpolation

Interpolation lower bound: setup

[A. Buna, S. X. Liu, P. Rebeschini 2026]

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Idea: whiten by $\boldsymbol{\Sigma}^{-1/2}$ and rotate by an orthogonal $\boldsymbol{R}$, so that Bayes-classifier direction splits into one signal coordinate and $d-1$ nuisance coordinates

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Interpolation lower bound

[A. Buna, S. X. Liu, P. Rebeschini 2026]

Interpolation lower bound

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Margin of w:

$$m(\boldsymbol{w}) := \min_{1\le i\le n} \frac{Y_i \boldsymbol{X}_i^\top \boldsymbol{w}}{\|\boldsymbol{w}\|}$$

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Interpolation lower bound

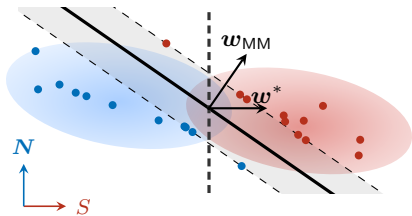
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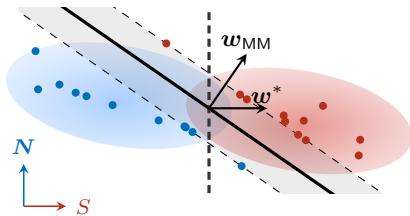
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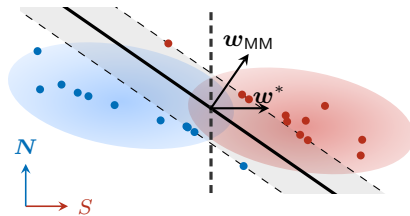
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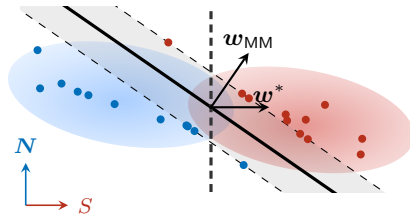
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- $d \gg n$: order $\frac{n}{d}$.

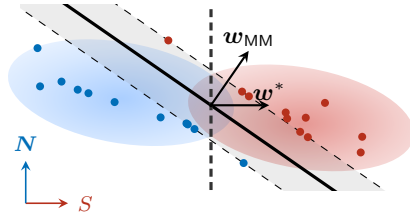
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where $\psi^2 + \|\boldsymbol{\nu}\|^2 = 1$, $\psi = \cos\angle_{\boldsymbol{\Sigma}}(\boldsymbol{w}, \boldsymbol{w}^*)$.



Linear interpolators: $\mathcal{I} := \{\boldsymbol{w} : \ m(\boldsymbol{w}) > 0\} \ni \boldsymbol{w}_{\mathrm{MM}}$.

Theorem 2 (informal). If $n \gtrsim \log(1/\delta)$, then with probability at least $1-\delta$, for every $\boldsymbol{w} \in \mathcal{I}$,

$$\mathcal{E}(\boldsymbol{w}) - \mathcal{E}(\boldsymbol{w}^*) \ \gtrsim \ (1-2p)\, \frac{n}{\big(\sqrt{n} + \sqrt{d-1} + \sqrt{\log(1/\delta)}\big)^2}\,.$$

- $d \gg n$: order $\frac{n}{d}$.
- $d \asymp n \log n$: order $\frac{1}{\log n}$, vs. early-stopping rates $n^{-\frac{2a}{2a+1}}$ and $\frac{\log n}{n}$.

Conclusion

Early stopping beyond regression

Early stopping beyond regression

	Least-squares regression	Linear classification

Early stopping beyond regression

	Least-squares regression	Linear classification
Interpolation		
Early stopping		

Early stopping beyond regression

		Least-squares regression	Linear classification
Interpolation			
Early stopping	**Implicit regularisation**		
	Statistical optimality		

Early stopping beyond regression

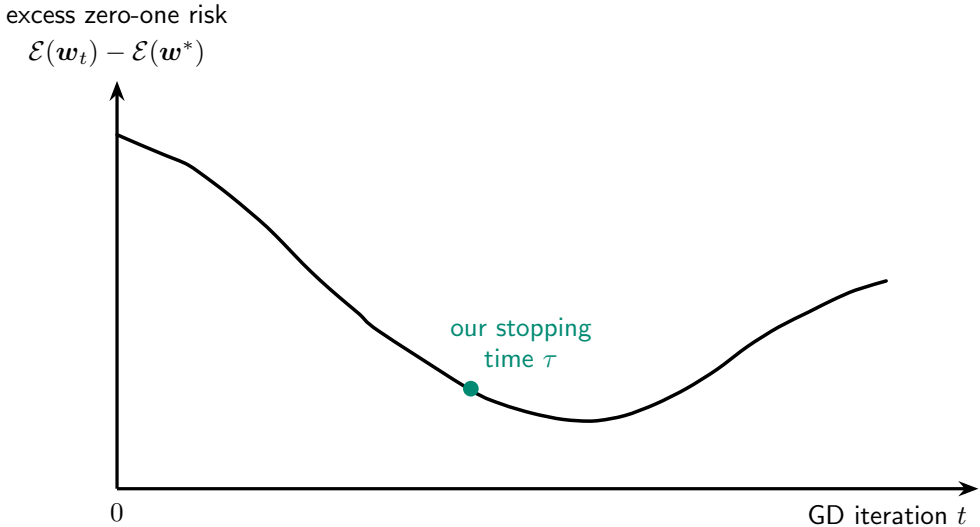
		Least-squares regression	Linear classification
Interpolation		Bartlett et al. [2020], Belkin et al. [2020], Hastie et al. [2022], Muthukumar et al. [2020], Tsigler and Bartlett [2023], Liang and Rakhlin [2020], Koehler et al. [2021], ...	
Early stopping	**Implicit regularisation**	Yao et al. [2007], Suggala et al. [2018], Ali et al. [2019], Wu et al. [2025a], ...	
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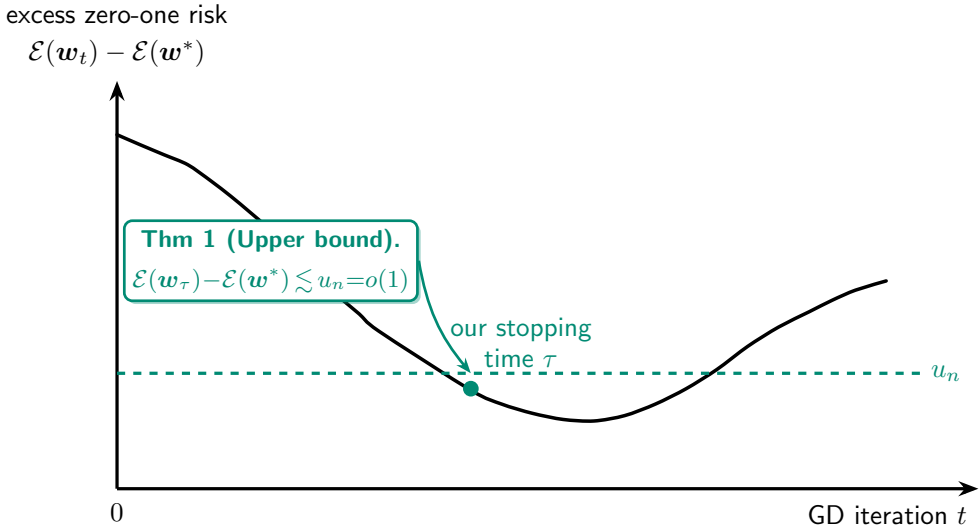
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Part 1: Statistical optimality of early stopping

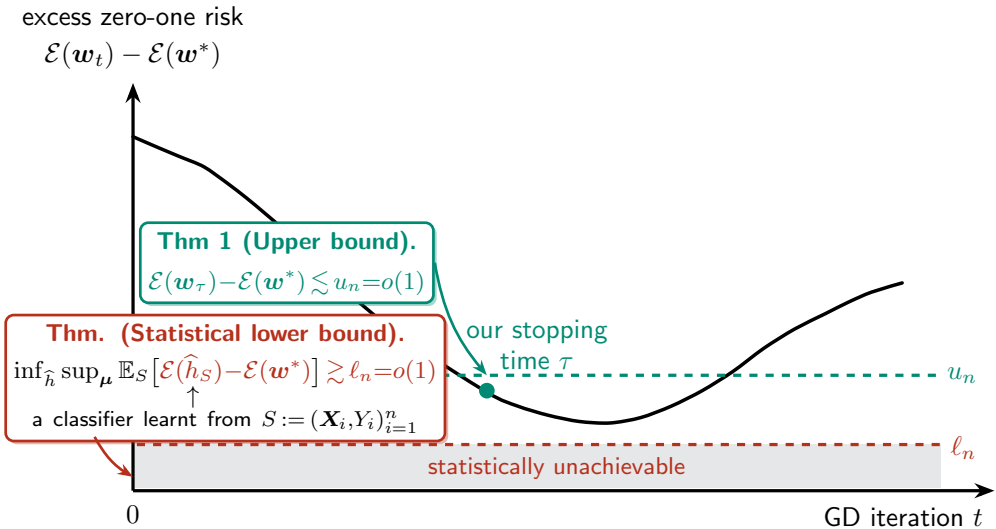
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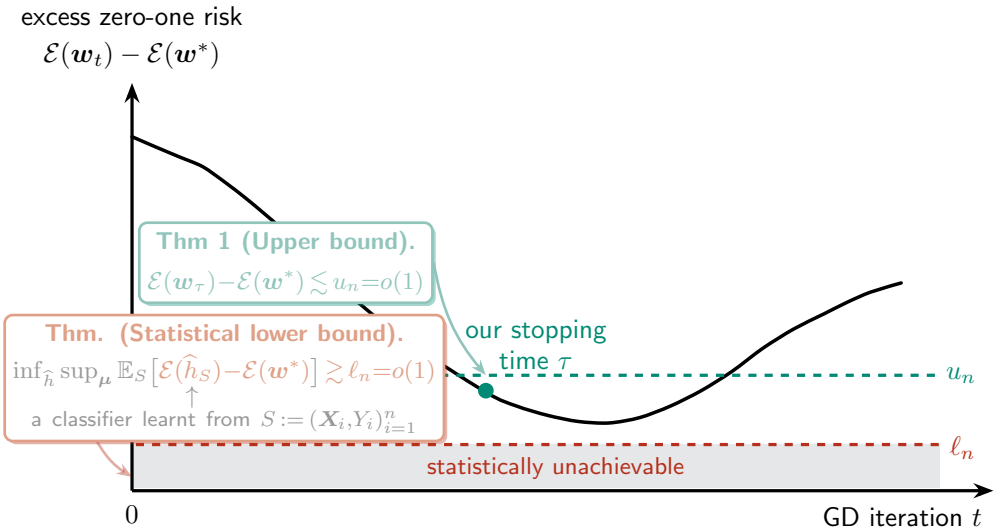
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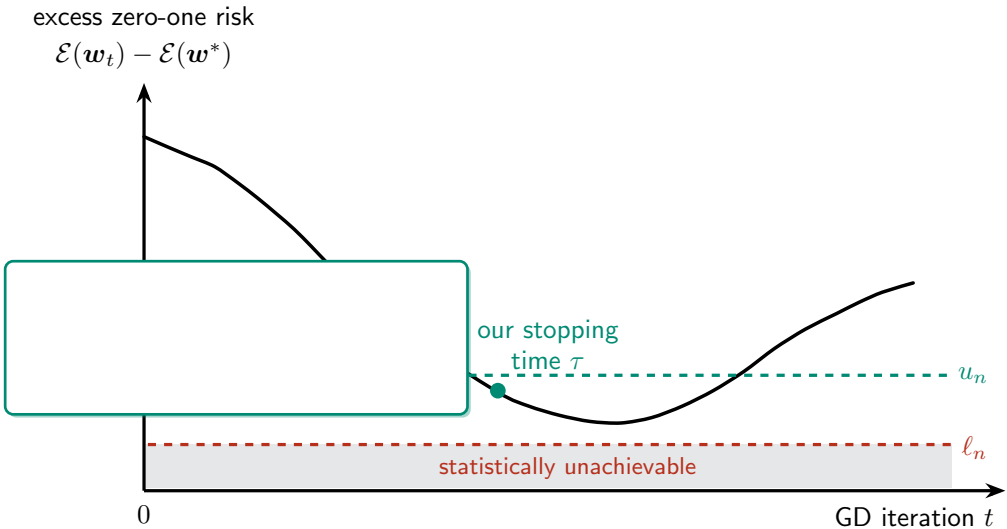
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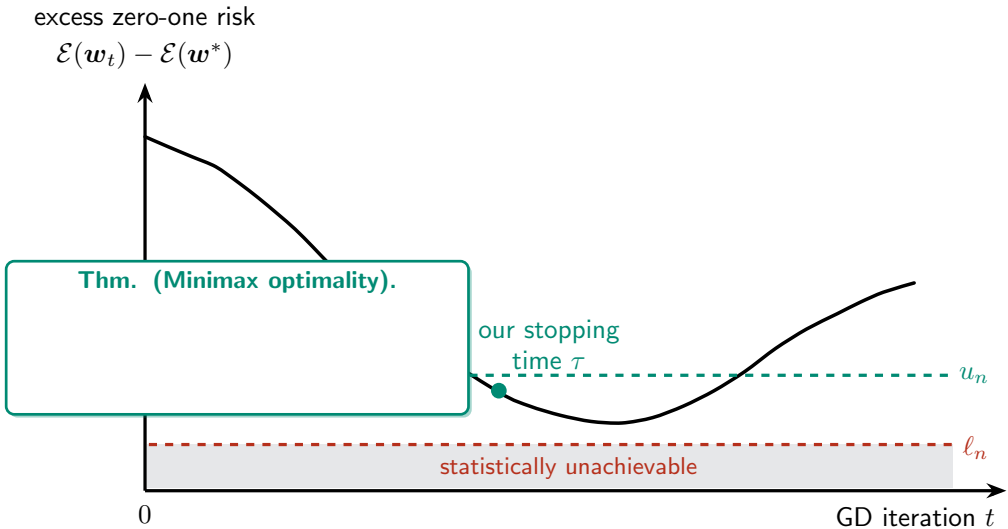
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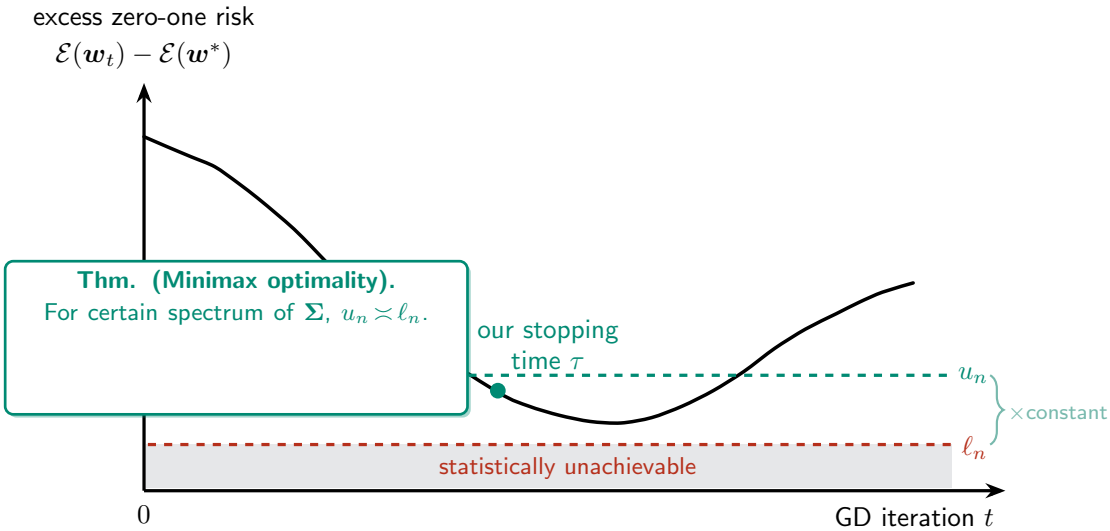
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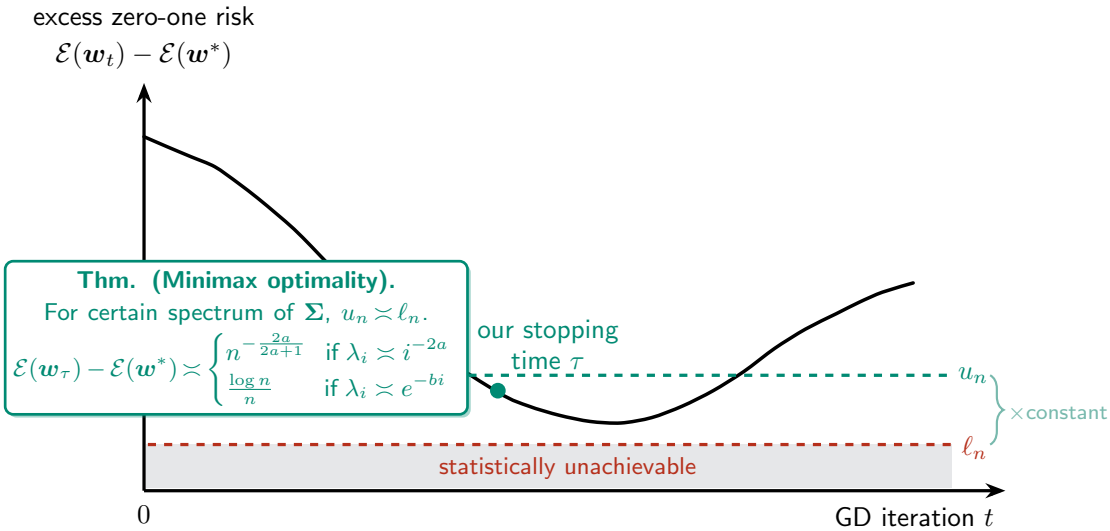
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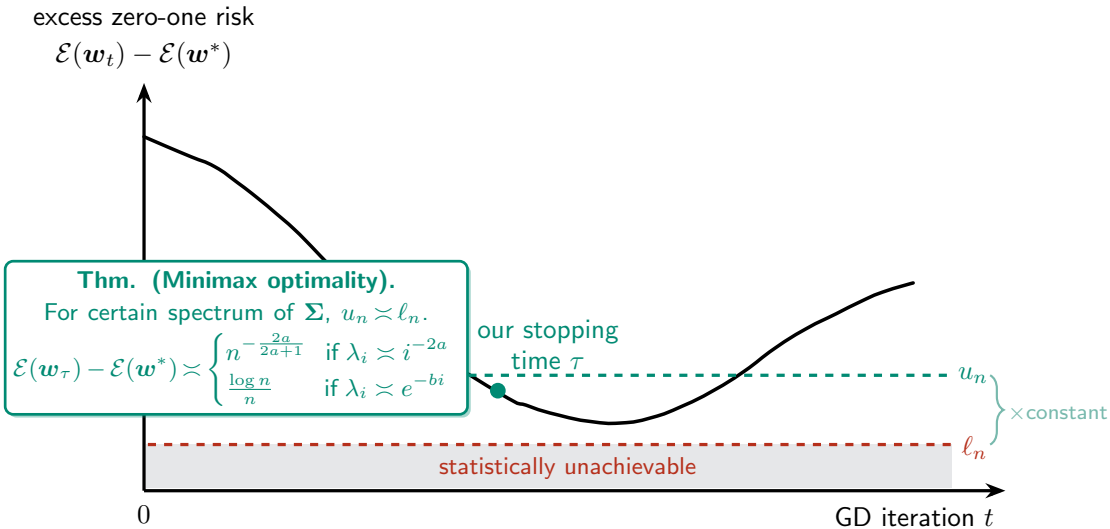
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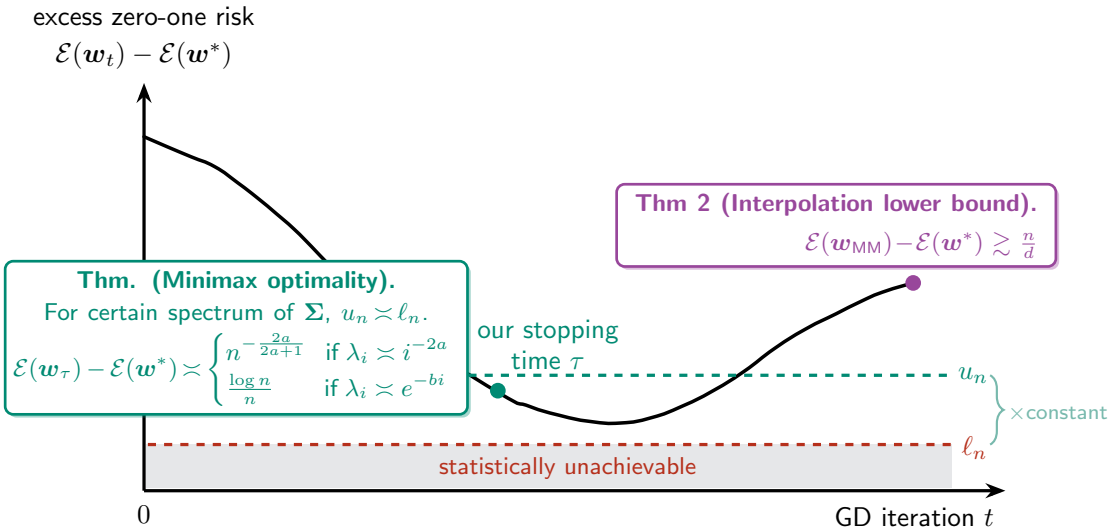
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Part 2: Statistical suboptimality of interpolation

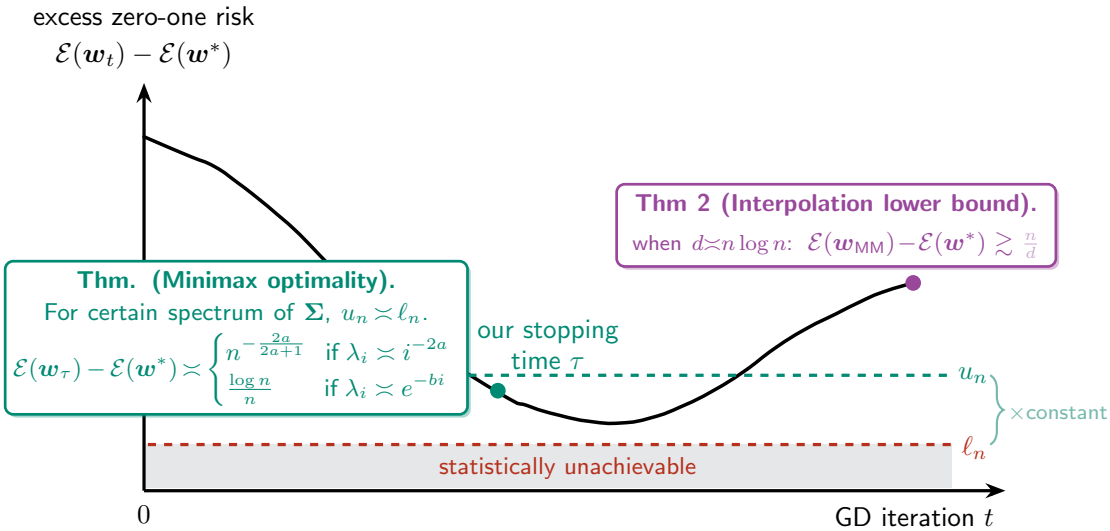
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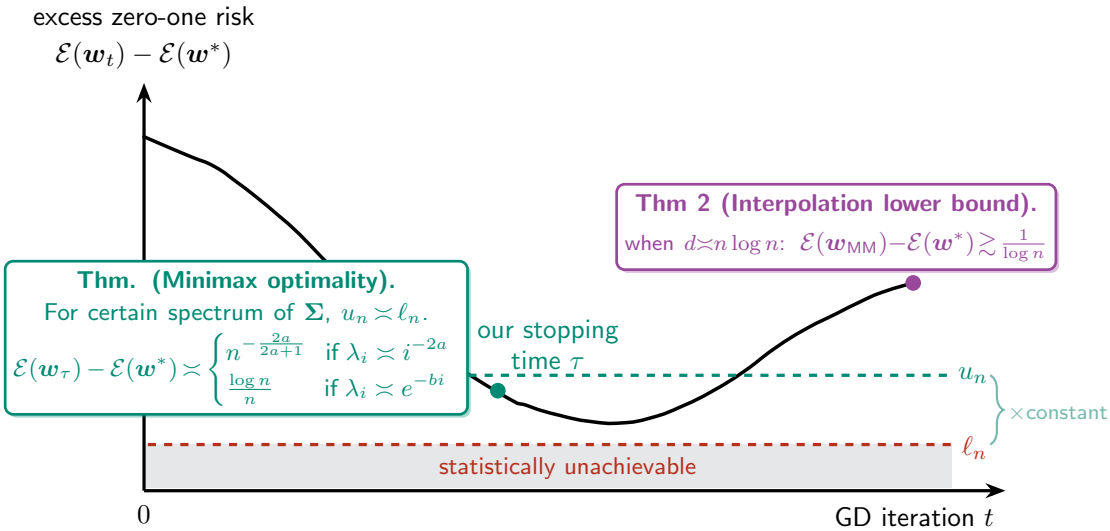
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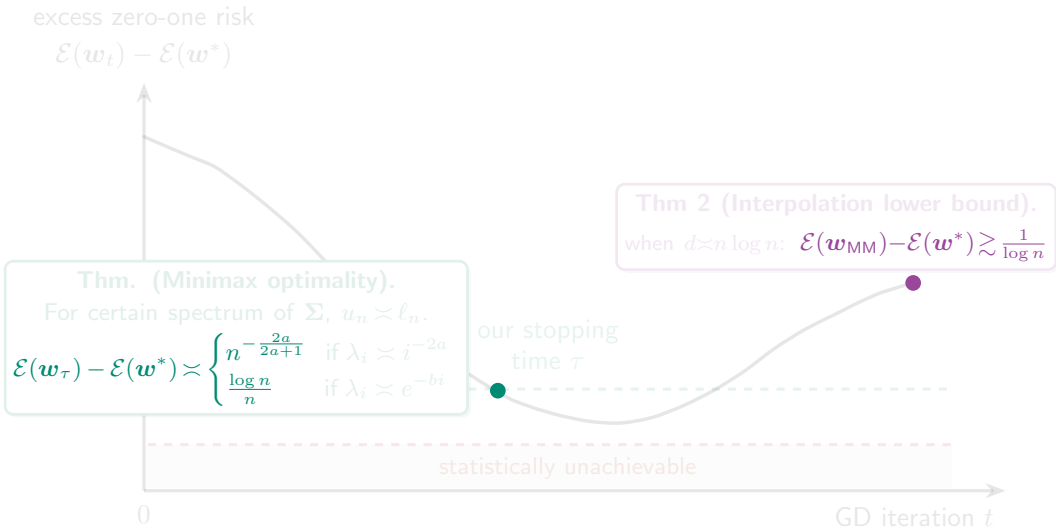
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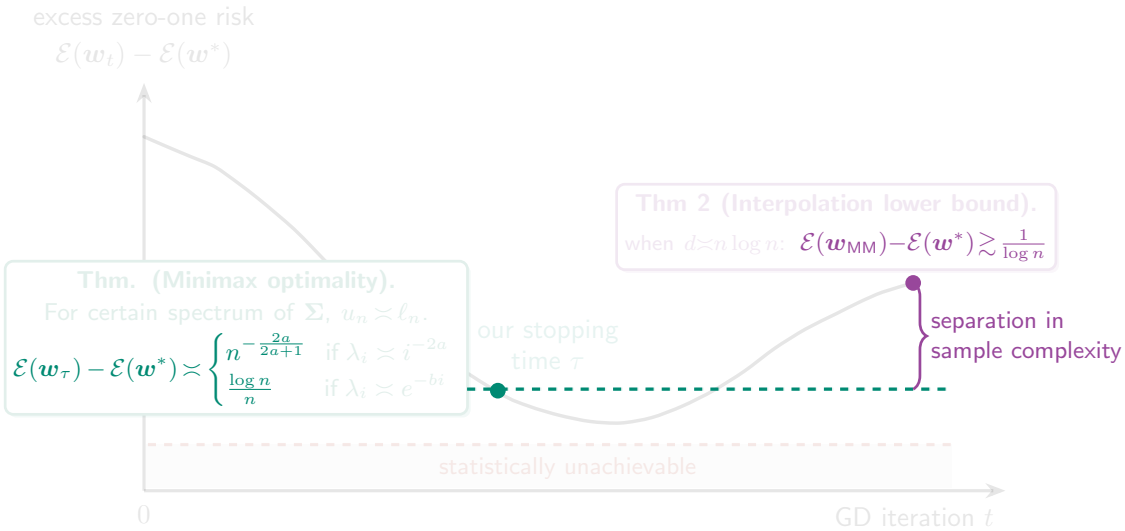
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